

Chapter 6: Space Physics

(1) Earth & The Solar System

a) Intro

b) The Earth

c) The solar system

$$V = \frac{2\pi r}{T}$$

(2) Stars & The Universe

$$1 \text{ light year} = 9.5 \times 10^{15} \text{ m}$$

a) Sun as a star

$$1 \text{ Solar mass} = 2 \times 10^{30} \text{ kg}$$

b) Stars Life Cycle

c) The expanding Universe.

$$H_0 = \frac{v}{d}$$

① a) Intro

Planets in the Solar System

Sun → Mercury → Venus → Earth → Jupiter
↓
Neptune ← Uranus ← Saturn

M V E M J S U N

- Pluto no longer considered a planet (but a dwarf),
planet
because scientist discover Eris (same size as Pluto)

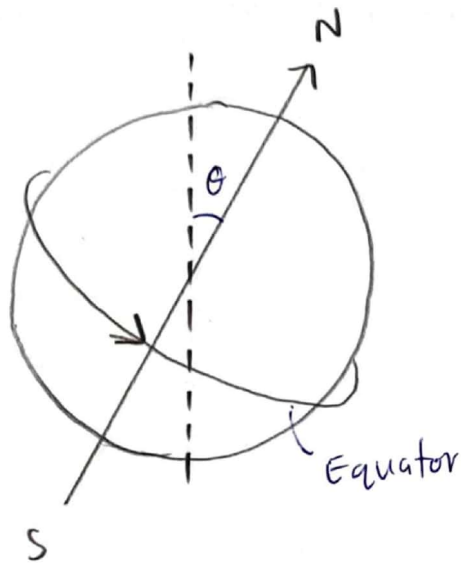
- Condition to be a planet!

1) Orbit the Sun

2) Have enough mass that its gravity pull
into a sphere.

3) Have large enough gravity (g) to clear
away any other objects of a
similar size near its orbit

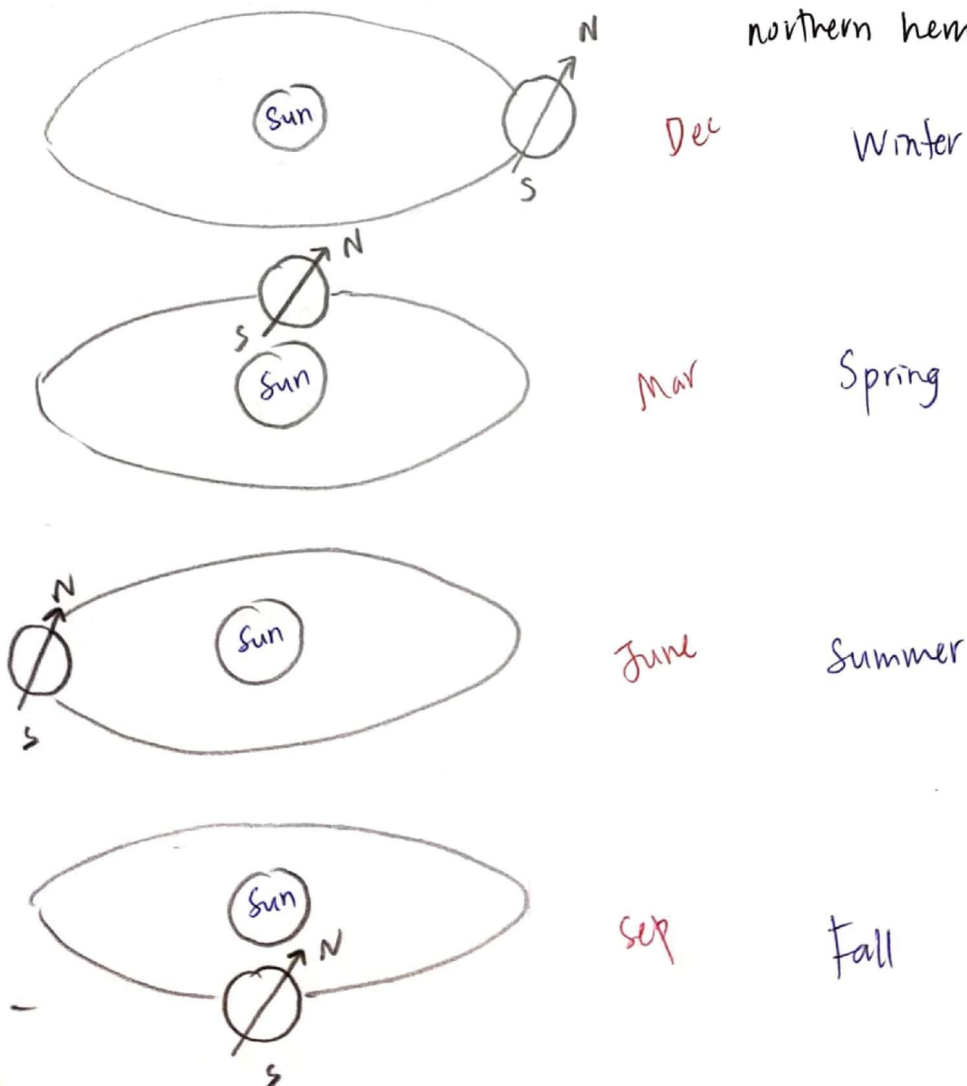
① b) The Earth



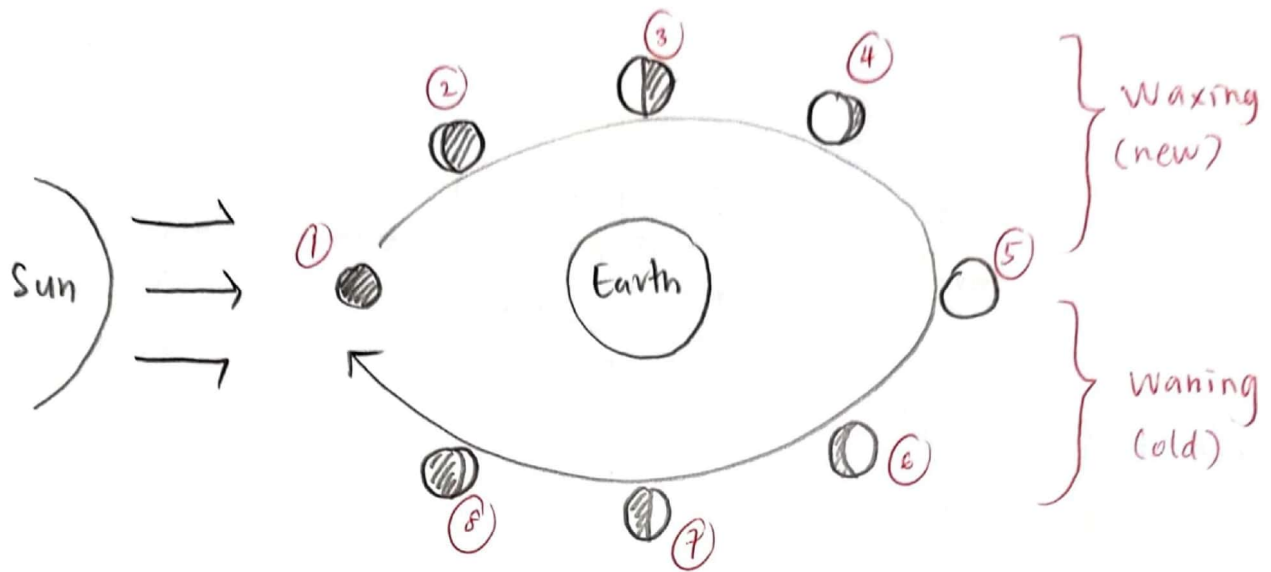
- Earth is tilted at an angle 23.5°
- Earth rotates counter-clockwise and causes day and night. (24 h)
- The 23.5° tilt causes changes of season which does not occur at the equator.

Earth orbits the Sun in 365 days (1 year)

Season at the northern hemisphere



Phases of the moon causes change in month

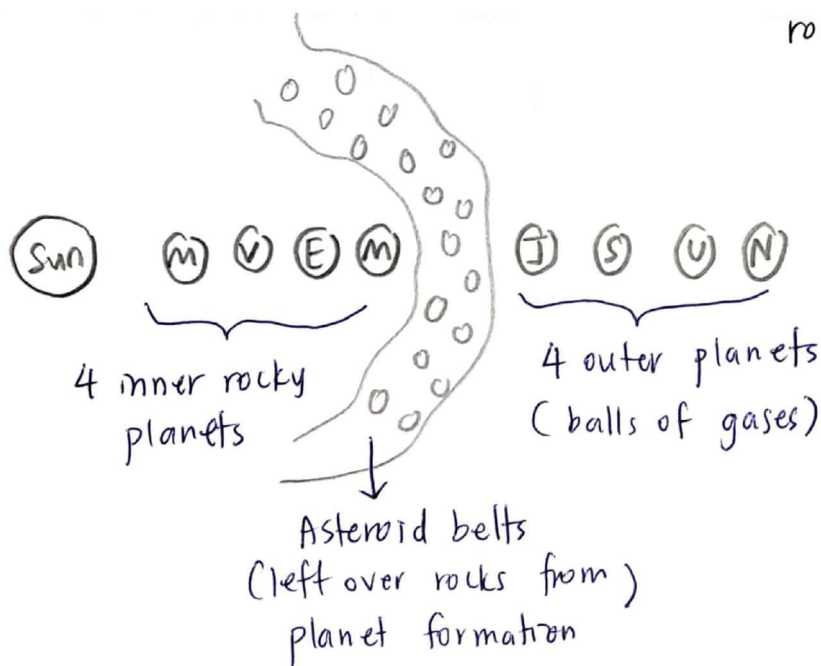
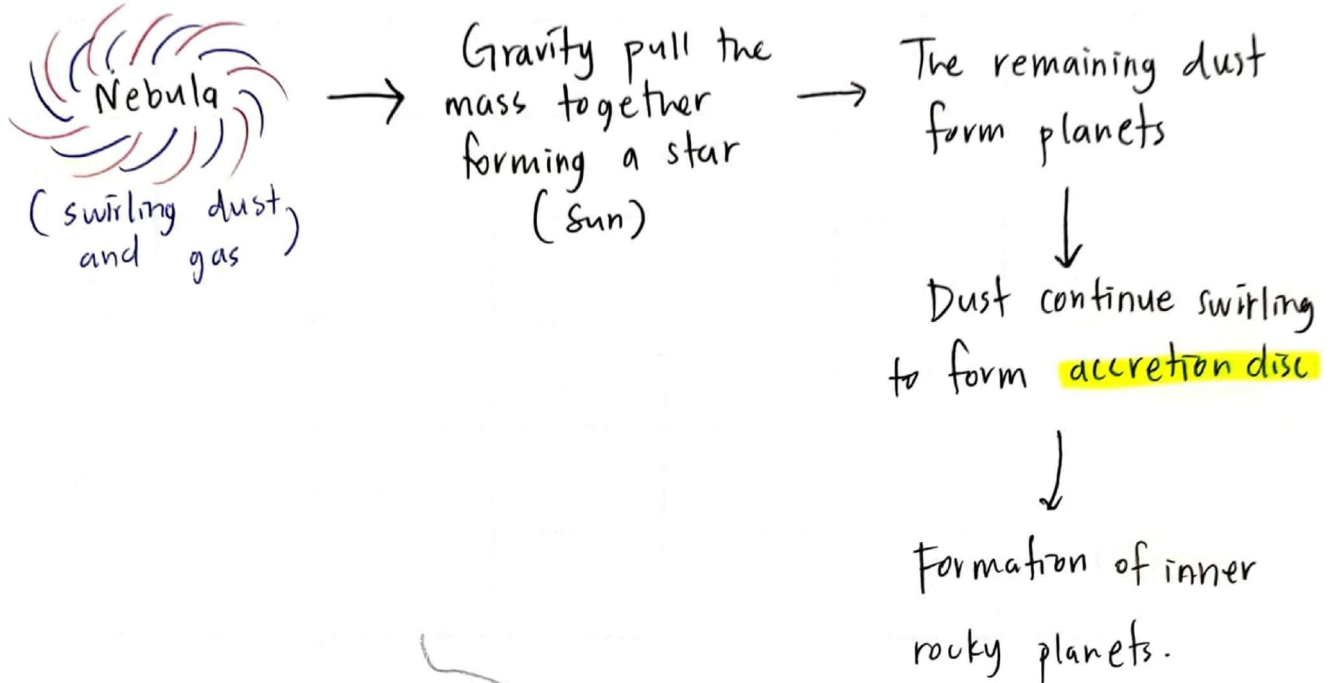


- (1) New moon
- (2) Waxing crescent
- (3) First quarter
- (4) Waxing gibbous (convex)
- (5) Full moon
- (6) Waning gibbous (convex)
- (7) Third quarter
- (8) Waning crescent

1 complete moon phase = 27.5 days

①c) The solar System

Formation of planets.



Asteroids

- Space rocks smaller than planets

meteoroids

- small space rocks that burns when it enters earth atmosphere

Table of info about planets in the solar system

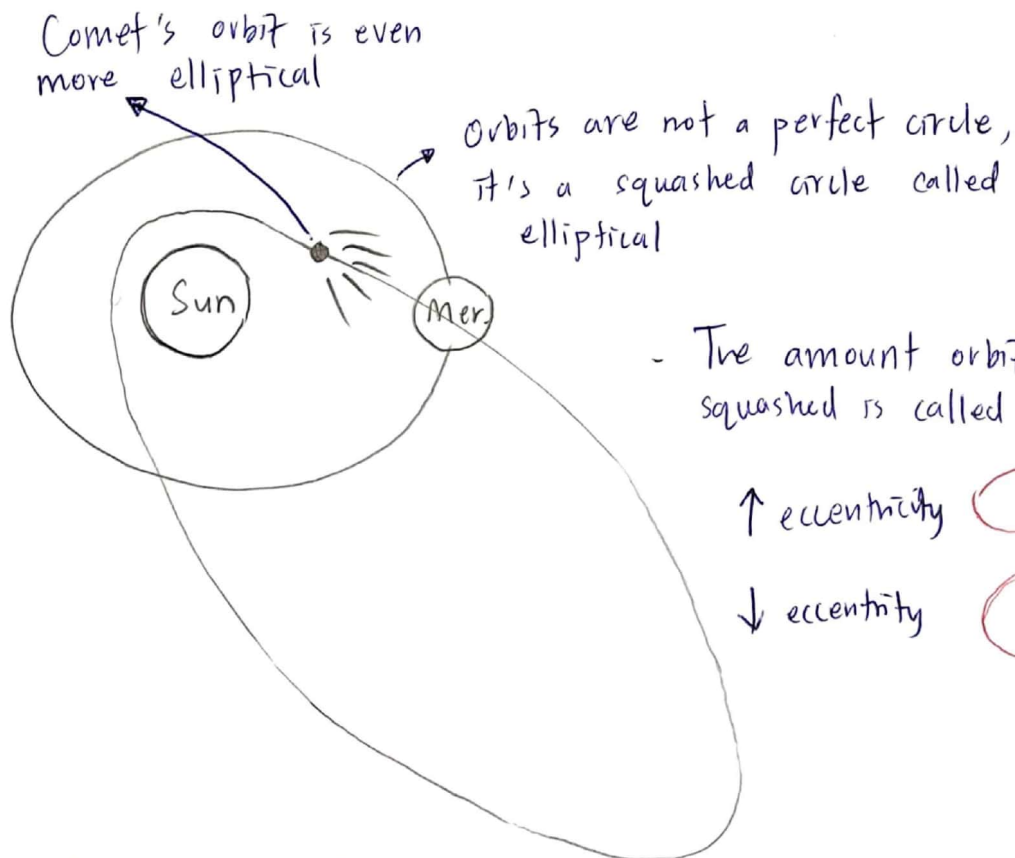
Planet	(10^6 km) orbital distance	(years) orbital duration	Density (kg/m^3)	Surface Temp ($^{\circ}\text{C}$)	Gravity N/kg	No. of moons
Mercury	58	0.2	5500	-18 to 460	4	0
Venus	108	0.6	5200	470	9	0
Earth	150	1	5500	-8 to 58	10	1
Mars	228	1.9	4000	-8 to -5	4	2
Jupiter	778	12	1300	15 to 20	26	16
Saturn	1427	30	700	-140	11	20
Uranus	2870	84	1300	-200	11	15
Neptune	4497	165	1700	-220	12	8

Orbits and Energy

- Other than planets, moon and asteroids, comets also orbit the Sun.

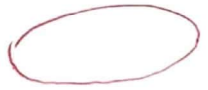
Comet? - Frozen leftovers from formation of solar system composed of dust, rocks and ices.

(Just like asteroids, but COLD)



- The amount orbit is squashed is called eccentricity

↑ eccentricity

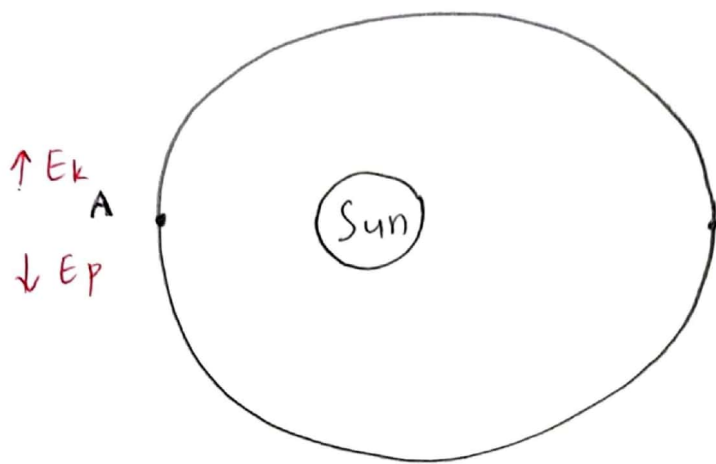


↓ eccentricity



- Comet Halley's orbit is much more eccentric than other planets in the solar system.
- Halley's orbital period is 75 years.

(Last seen (1986) → Back online (2061))



- Sun is not exactly in the middle of a planet's orbit.

- This caused planets to travel at different speed depending on its distance from the Sun.

- At A, planet is closer to the Sun, so it will move the fastest because gravitational pull from the Sun is greatest at this point.

So, At A
- fastest
- closest

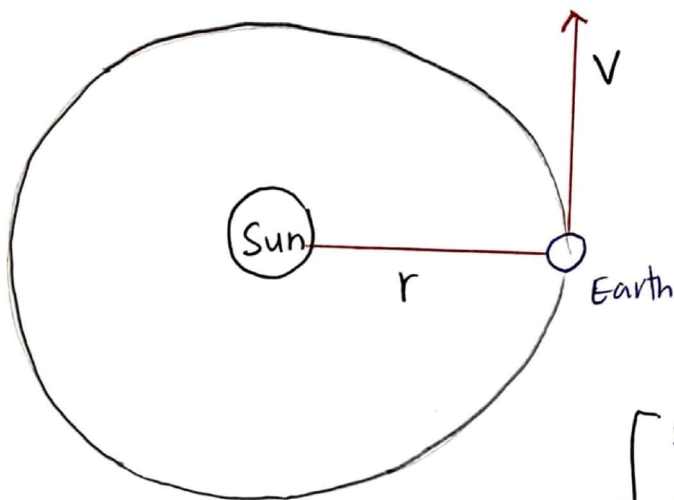
- So at A, the planet will have highest kinetic energy, and potential energy at its lowest

At B
- slowest
- furthest

- At B, since the planet is at a farthest point, Sun's gravitational pull will be at its lowest so the planet will have highest potential energy and lowest kinetic energy.

Orbital Speed, v

- Speed of planet in orbit round a star
- Even though orbit is not perfectly circle (ellipse), we take its average orbital radius, r (refer table of info)



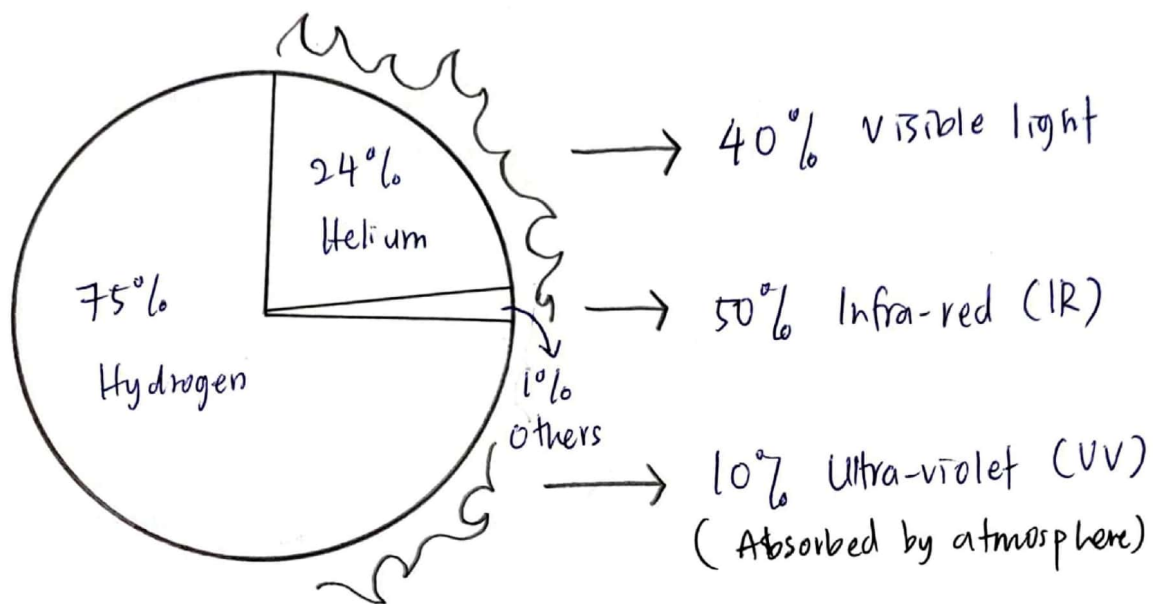
Assume the orbits are circular

using $s = \frac{d}{t}$

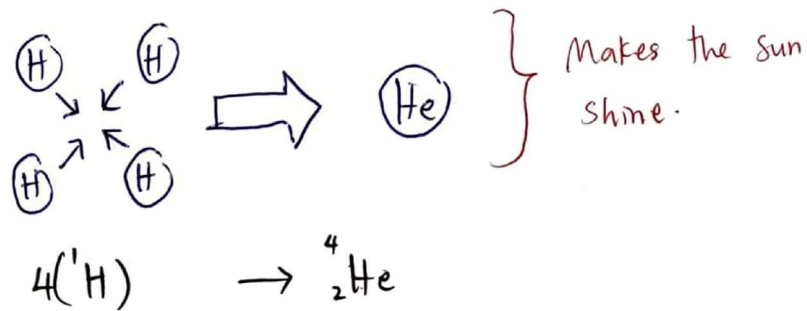
$\left\{ \begin{array}{l} s = \text{speed} \\ d = \text{average orbital distance} \\ t = \text{orbital period} \end{array} \right.$

$$V = \frac{2\pi r}{T}$$

(2).a) The Sun as a Star



- Sun is a medium mass (stable) star
- Sun has a mass of 2×10^{30} kg (1 solar mass)
 - ↳ Make up 99.86% of solar system's mass.
- Powered by Nuclear fusion (in the sun's core)

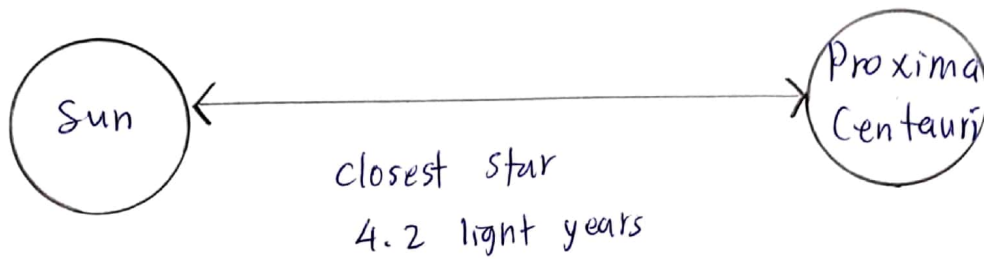


Sun's core (15 000 000 K)

- Hydrogen exists as plasma
(positive ions and electrons)
- A form of ionized gas

Surface of the Sun (Photosphere)
(5800 K)

② b) Stars



1 light year = how many km? It's a distance travelled by light for 1 Earth year.

Using $s = \frac{d}{t}$

s = speed of light (3×10^8 m/s)

t = one year ($1 \text{ y} \times 365 \text{ d} \times 24 \text{ h} \times 60 \text{ min} \times 60 \text{ s}$)

d = distance travelled by light

$$d = s \times t$$

$$d = (3 \times 10^8) \times (1 \times 365 \times 24 \times 60 \times 60)$$

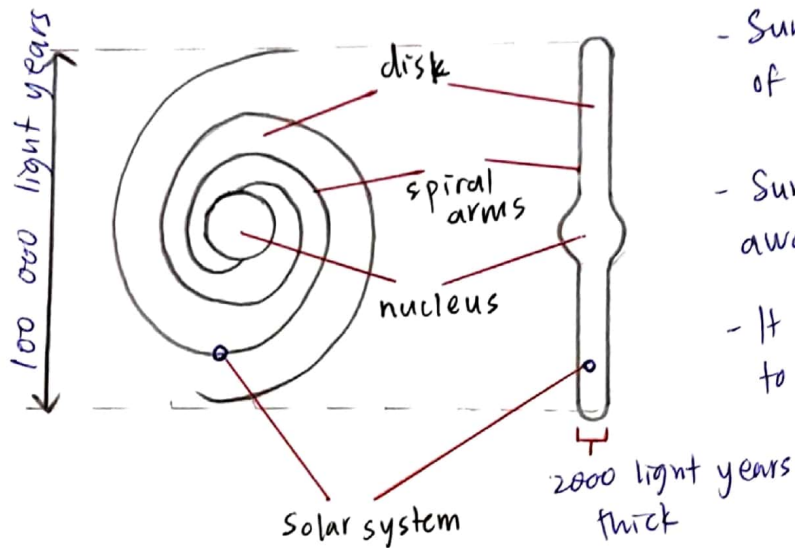
$$d = 9.5 \times 10^{15} \text{ m} \approx \boxed{9.5 \times 10^{12} \text{ km}}$$

is the distance travelled by light for 1 year.

$$\text{So, } 4.2 \text{ light years} \times 9.5 \times 10^{12} \text{ km}$$

$$= 3.99 \times 10^{13} \text{ km (Sun} \rightarrow \text{Proxima Centauri)}$$

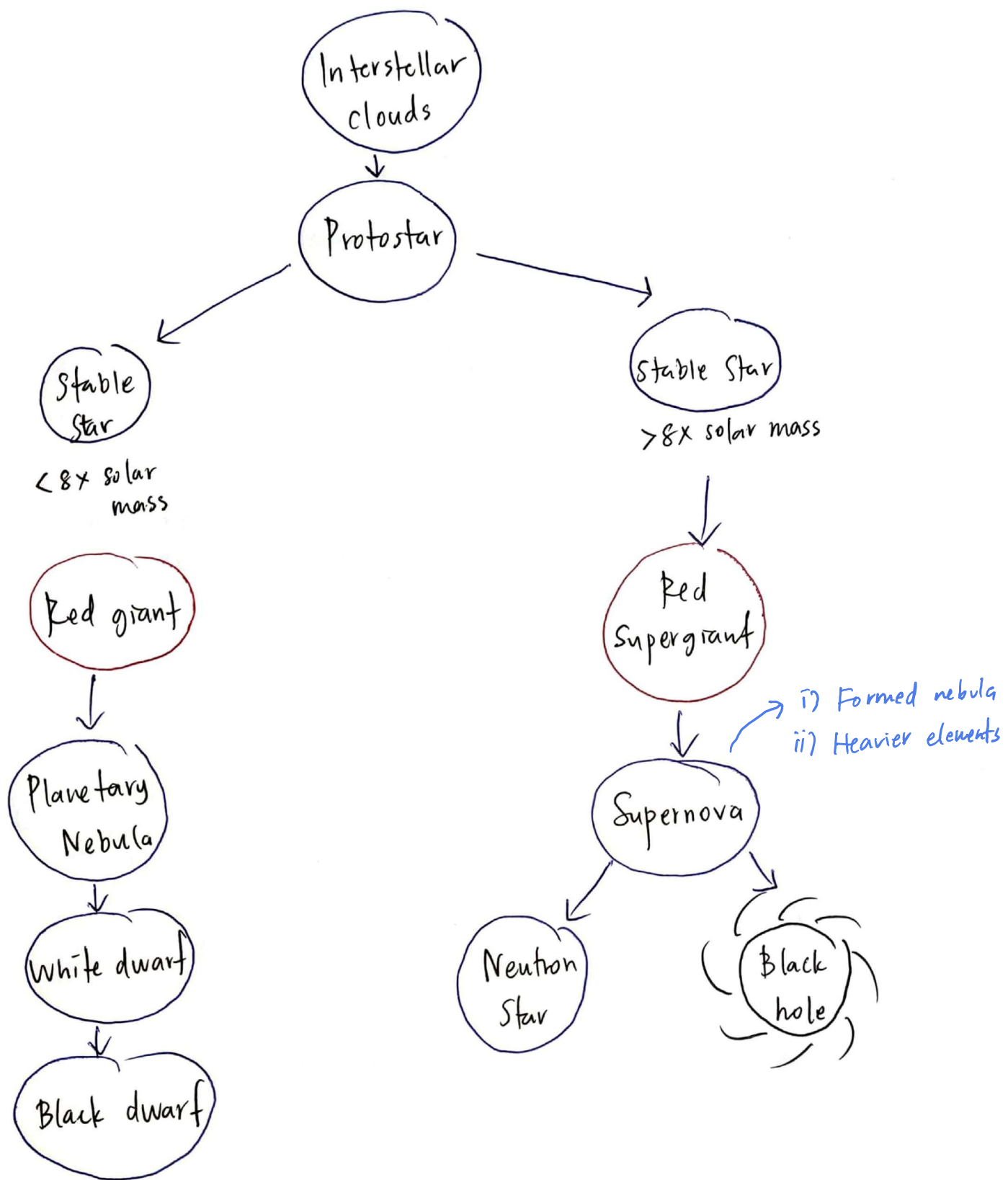
Our Galaxy (Milky Way)



- Sun is one of 200 billions of stars in Milky Way
- Sun is 30 000 light years away from nucleus
- It takes 225 million years to travel once across the galaxy (1 cosmic year)

- Closest galaxy is Andromeda (2.5 million light years)
- Canis Major Dwarf Galaxy (25 000 light years)

Life cycle of a Star



Interstellar clouds

L Gas and dust that contain hydrogen called molecular clouds

Protostar

L Young star that is still gathering mass from its parent molecular cloud.

L molecular clouds must be

- ① Cold } for star formation
- ② dense }

Orion Nebula (in Milky Way)

- visible star formation (1350 light years) away

Stable Star Formation

① Molecular & dust collapse caused by gravitational attraction



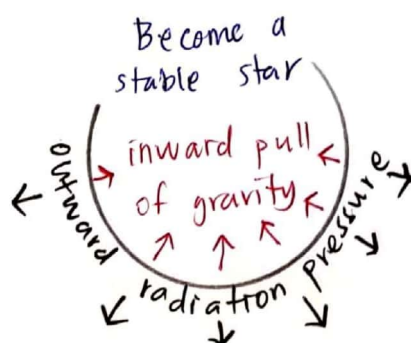
GPE → kinetic energy
Thermal energy



Spinning sphere of super hot gas (Protostar)



Growing by pulling more materials from molecular clouds



(Star $< 8 \times$ solar mass)

Red giants

- Stars run out of hydrogen
- Become bigger star with cooler surface.

White dwarf

- Core of red giants collapse into white dwarf
- cannot exceed 1.4 solar masses
- radius < 1000 km
- 1% of its original radius
- Hot surface but not hot enough for nuclear fusion
- Cool down to become black dwarf

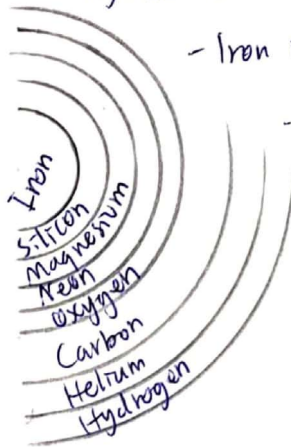
Planetary nebula

- A bubble of gas surrounding a white dwarf star that used to be the outer shell of a red giant from which it collapsed.

(Star $> 8x$ solar mass)

Red supergiants.

- Core of massive stars get so hot that the nuclei of heavier elements can fuse.
- Outer layer of core hot enough for fusion of lighter elements



- Iron is the heaviest

- Outer shell expands into red supergiants.

Supernova

- As fuel run out, the star collapse and explode.
- During supernova explosion, the energy is enough to fuse elements heavier than iron
- The elements are scattered in space as nebula

Neutron star

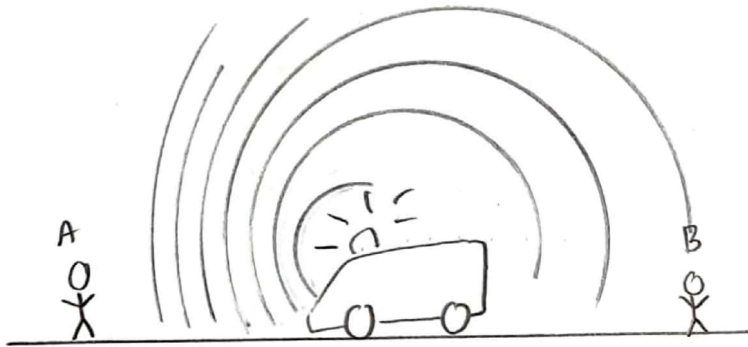
- Form from a collapse of star with a core ... of $3x$ solar mass
- Collapse of heavier core will form black hole.

②-c) The Expanding Universe

Theory based on astronomical observations & ideas such as ① Red-shift
② Dark energy

Describe redshift as an increase in the observed wavelength of electromagnetic radiation emitted from receding stars and galaxies

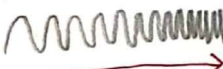
The Doppler Effect



The further the van, the bigger the gap of sound wave that reach Person B

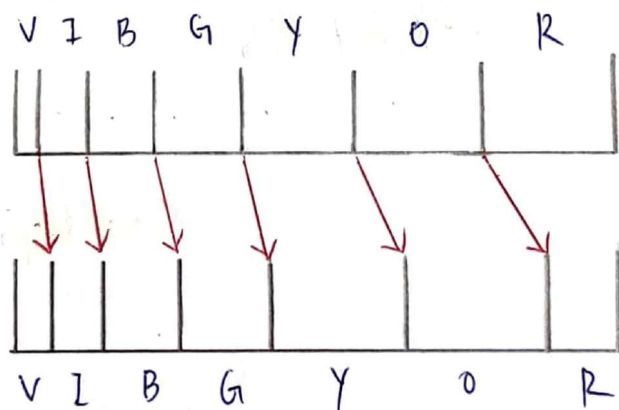
- As van move closer to Person A, the wave compressed ahead (higher pitch sound) and the wave behind the van are redshifted behind (lower pitch sound) heard by Person B.

Light spectrum (Spectroscopy - study of light)

ROYGBIV

Higher frequency.

- Absorption spectrum
- wavelength of light missing from sunlight as the cool gas in Sun's atmosphere absorbs them.

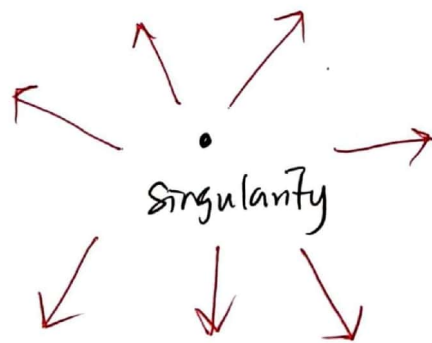
Light from Sun



Light from distant galaxy

↑ redshift, faster the galaxy move away

- Comparing light from sun & stars of distant galaxy, astronomers discovered that the spectrum are redshifted
 - ↳ shifted towards the end of spectrum (just like The Doppler Effect).
- This means the wave from distant star is stretched out and wavelength is increased.
- This is clue that shows the stars from distant galaxy move away from us.
(Universe is expanding).
- The universe (space, time, matter, energy) was created at a single point 13.8 billion years ago.



Explode in the
Big Bang and
continue expanding.

Hubble's Law

- The speed of galaxies is directly proportional to the amount of redshift.

$$v = H_0 d$$

$$H_0 = \frac{v}{d}$$

H_0 = Hubble constant

v = speed of galaxy moving away from Earth

d = distance of galaxy from Earth

The current estimate for H_0 is 2.2×10^{-18} per second.

H_0^{-1} is known as Hubble time.

- We use it to find the age of the universe

$$t_{\text{universe}} = \frac{1}{H_0}$$

$$= \frac{1}{2.2 \times 10^{-18}} = 4.5 \times 10^{17} \text{ s}$$

$$= 14.4 \times 10^9 \text{ years.}$$

Cosmic Microwave Background Radiation (CMBR)

- This comes from all direction in space
- Temperature of about -270°C
- CMBR is the remains of thermal energy from the Big Bang, spread thinly across the universe.

Dark Energy

- Caused the Universe to expand faster
- Made up 68% of the Universe

Dark Matter

- Invisible mass that cause stars to rotate faster
- Has gravity that affects objects nearby.
- Made up 27 % of the Universe

Only 5% of the Universe is made of matter (stars, planets, people).